

Eva Dimitrova

443-430-4198 | Columbia, MD | edimitro@terpmail.umd.edu | linkedin.com/in/eva-dimitrova | evadimitrova.github.io

EDUCATION

University of Maryland, College Park

College Park, MD

Bachelor of Science in Computer Engineering, GPA: 3.6/4.0

Expected May 2028

WORK EXPERIENCE

Software Engineer Intern

May 2026 – August 2026

Liberty Mutual Insurance

Portsmouth, NH

- Drove the team's first IBM ODM-to-microservice migration as the sole contributor, establishing a repeatable approach to be used for 20+ future service migrations.
- Developed a Java Spring Boot service to replace a legacy IBM ODM system handling 40,000+ daily requests, cutting latency by ~25% and reducing developer onboarding time from 3 weeks to 3 days.
- Cut deployment time from 3 weeks to minutes via Docker/Kubernetes on AWS and a GitHub Actions CI/CD pipeline with 50+ automated tests.
- Enabled ~\$453K in projected licensing savings across 20+ future migrations by building a multithreaded, OAuth2-secured Compare API tool that validates Spring Boot responses against legacy output and routes discrepancies to Splunk and Slack in real time.
- Built a Datadog dashboard with traffic and P95 latency widgets for the Spring Boot microservice, and configured automated health-check alerts in the CI/CD pipeline to flag service degradation.
- Prototyped a Slack-integrated employee recognition tool to replace a discontinued \$3K+/year SaaS subscription, using AWS Lambda, PostgreSQL, Node.js, and React.

Computer Engineering Intern

May 2025 – August 2025

Brown Deer Technology

Bel Air, MD

- Streamlined embedded testing workflows by developing a C-based libconfig parsing tool on Linux that generates JSON configurations and eliminates manual field entry.
- Enabled secure embedded deployments for mission-critical defense applications by integrating the seL4 microkernel on RISC-V architecture and debugging low-level IPC communication, kernel builds, and runtime issues.

Research Intern

May 2025 – August 2025

Maryland Information and Network Dynamics Lab

College Park, MD

- Reduced false positives by ~60% across 30+ building energy meters by engineering an automated data pipeline with Python that uses statistical modeling to detect anomalies in high-frequency sensor streams.
- Improved anomaly validation workflows for 15+ researchers by building visualization tools that surface anomalies directly on raw sensor data streams, eliminating manual graph inspection.

PROJECTS

Collaborative AI Chat Platform | Python, LangChain, PostgreSQL, pgvector

July 2026

- Won 1st place out of 21 teams (120+ participants) in Liberty Mutual's intern hackathon by building a collaborative AI chat platform that enables teams to share conversations and eliminate redundant AI queries.
- Owned the retrieval-augmented generation (RAG) pipeline using pgvector and LangChain to perform semantic similarity search over historical prompts and responses, retrieving relevant answers before invoking the LLM to reduce redundant AI costs.

YouTube Semantic Video Search System | Python, FastAPI, FAISS, JavaScript

January 2026

- Cut video lookup time by ~80% across 500+ liked videos using FAISS embedding search over titles/descriptions.
- Built a JavaScript search and browse interface, calling OAuth2-secured FastAPI endpoints that rank and filter results by similarity.
- Designed a FastAPI backend integrating FAISS vector search with YouTube metadata storage to support scalable semantic retrieval.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, SQL, Rust, OCaml, R, HTML

Frameworks & Tools: Spring Boot, AWS, Kubernetes, Docker, CI/CD, REST APIs, Git/GitHub, Linux, FastAPI, Flask, React, PyTorch, Datadog, Splunk